

POWER BI 30 DAYS COURESE

Day 4 - Append Data

Append Data means combining rows from two or more tables into one table.

Example:

Table 1

Customer
A
B

Table 2

Customer
C
D

After Append

Customer
A
B
C
D

Important:

- Columns should be the same or similar.
- Append adds rows (vertical combination).
- It does not add columns.

Append adds rows (vertical combination) means:

Rows = Horizontal lines in a table.

When we Append, data is added below the existing data.

Example

Table 1

Name	City
Ram	Delhi
Ravi	Mumbai

Table 2

Name	City
Amit	Pune
Neha	Jaipur

After Append

Name	City
Ram	Delhi
Ravi	Mumbai
Amit	Pune
Neha	Jaipur

New records are added downward (vertically).

It does not add columns means:

Append does not add new fields/columns side by side.

Adding columns is called Merge, not Append.

Example of Columns (Horizontal)

Name	City	Salary
------	------	--------

Here Salary is a new column.

Easy Interview Answer:

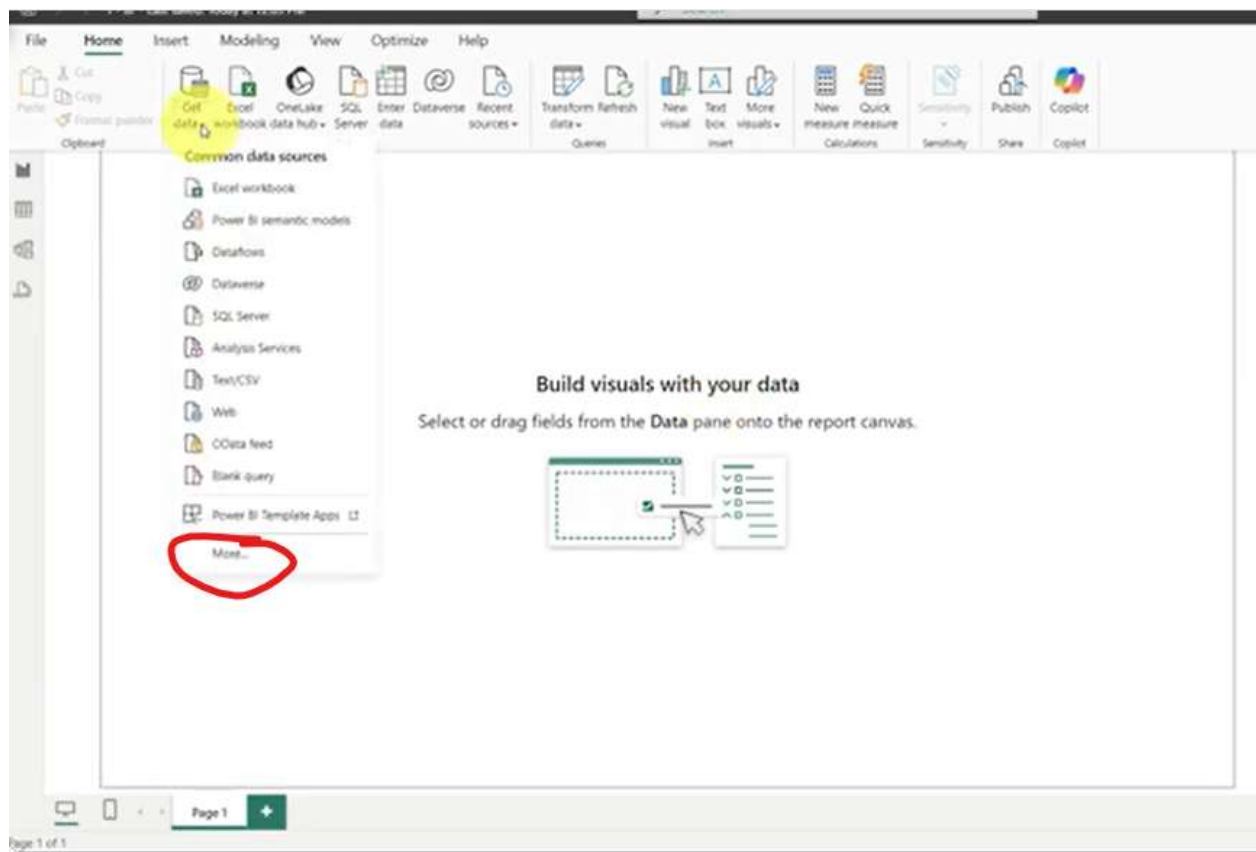
Append combines tables by adding rows below existing rows. It increases the number of records, not the number of columns.

or

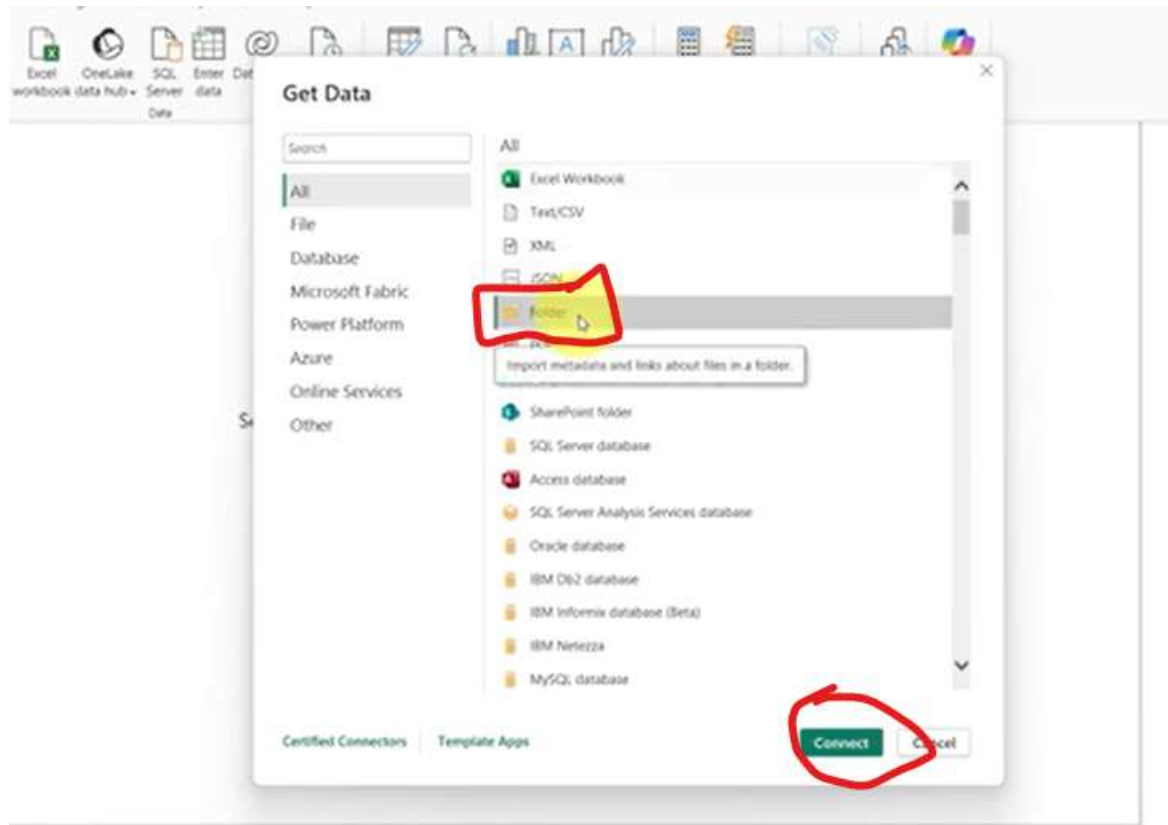
Append is used to combine tables by adding rows below existing data.
Merge is used to combine tables by adding columns side by side based on a common column.

Now, lets understand steps to append tables:

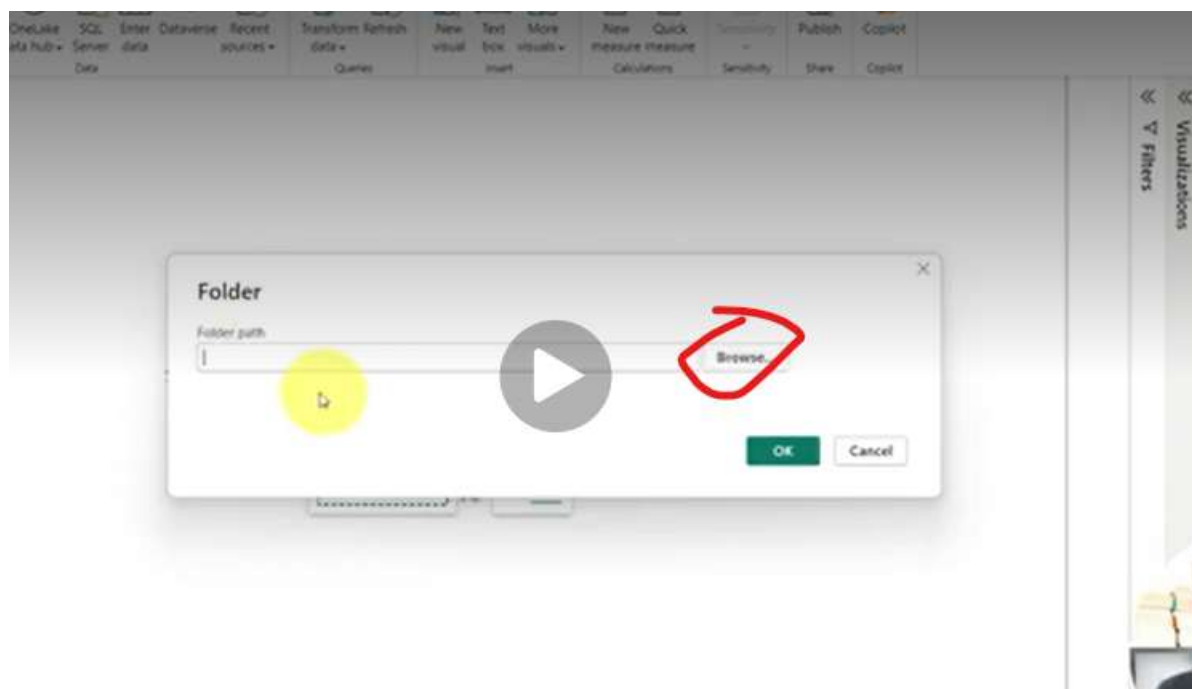
Step 1: click on get data →more

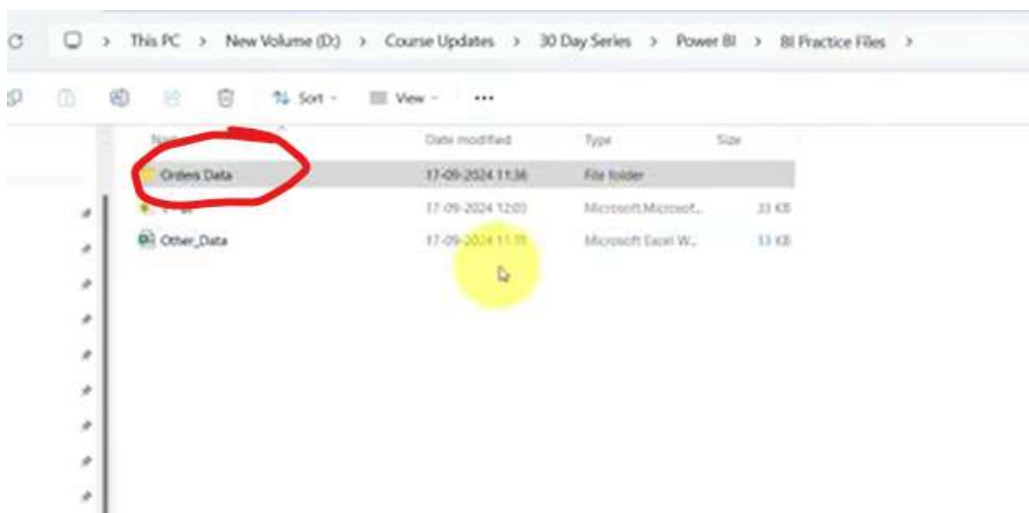
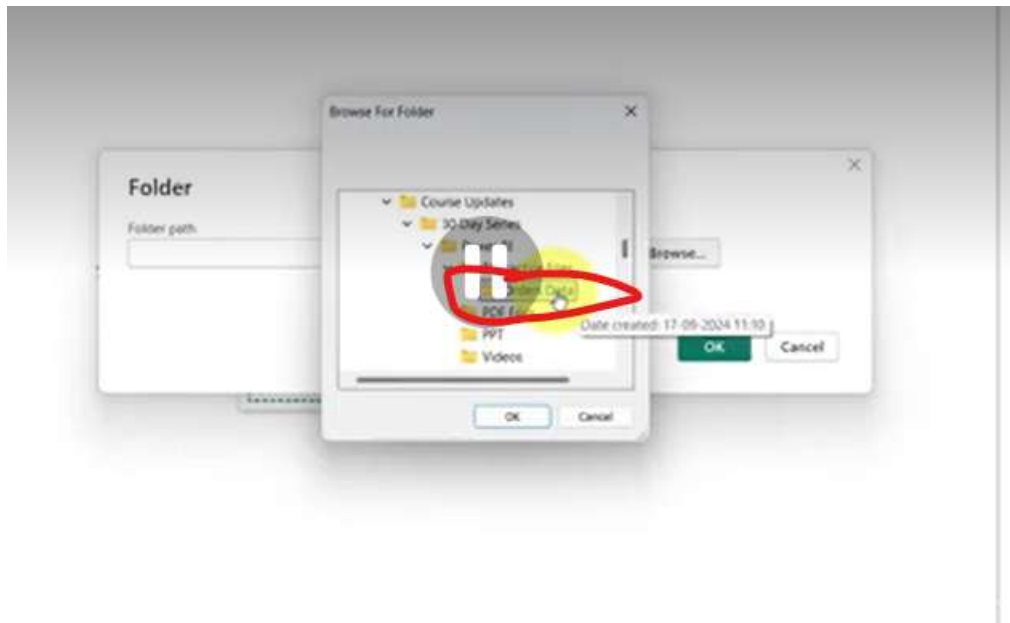


Step 2: a pop up box will come of get data in that select folder option and then click on connect.

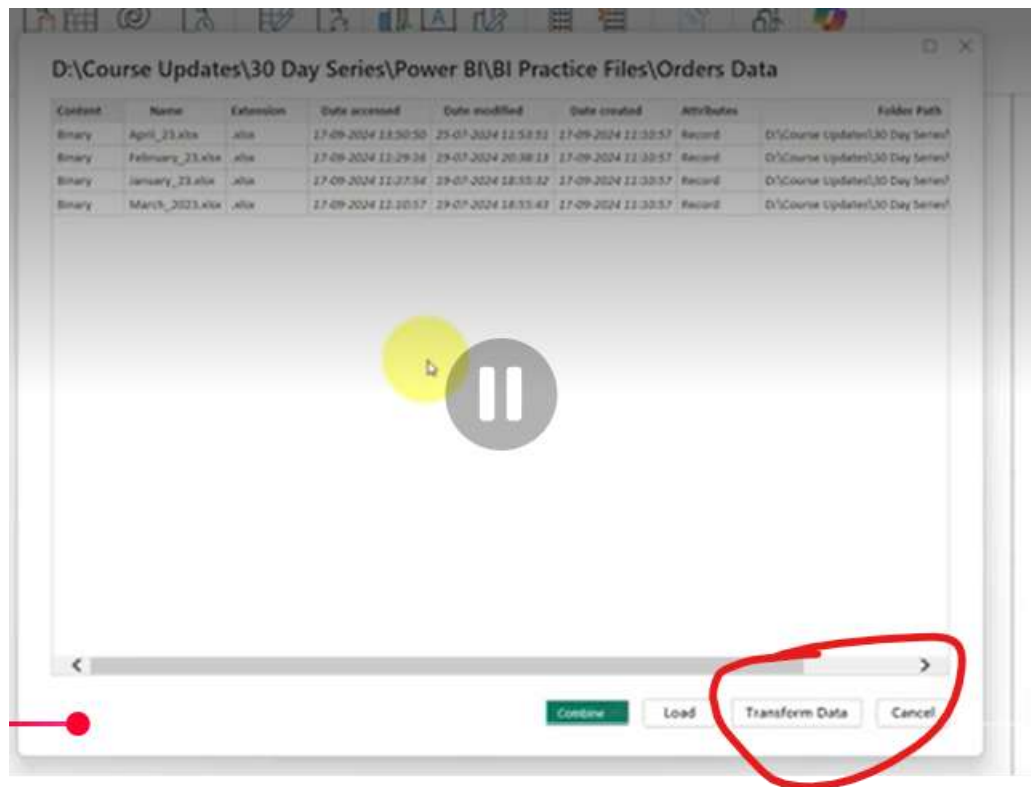


Step 3: a folder pop up box will come click on browse than select the Order Data folder than ok hence path will come of folder than click on ok.

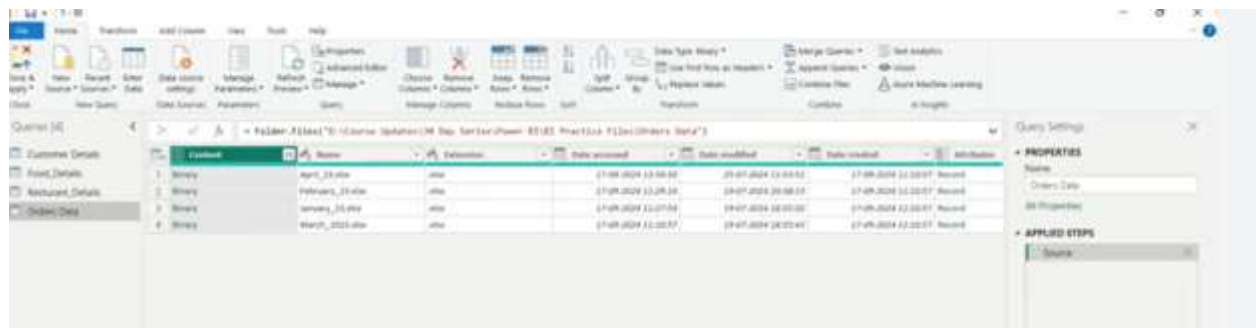




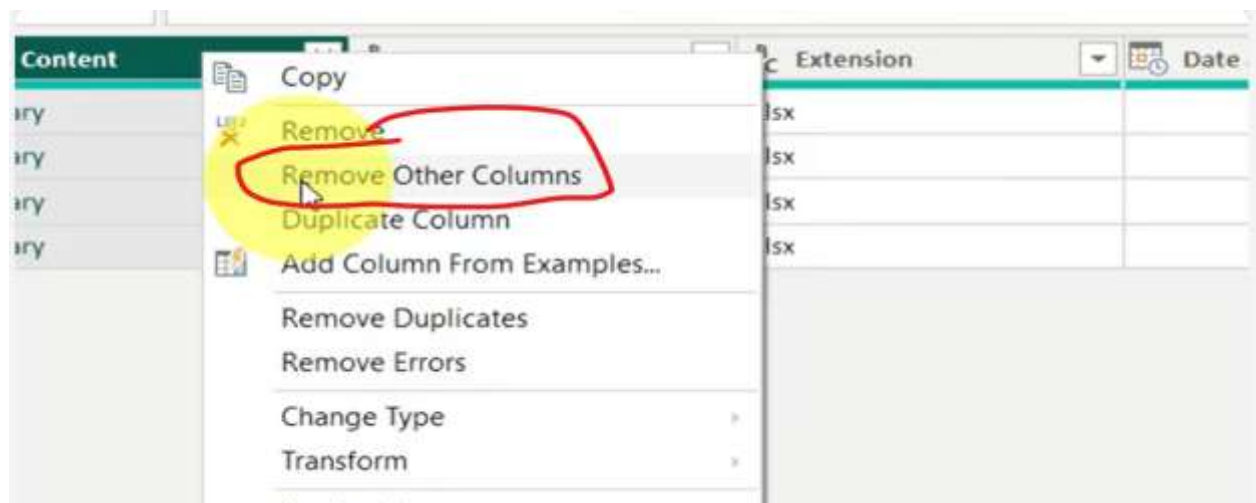
Step 4: folder is now open with binary data and files name other property .
Now, we will click on transform data .

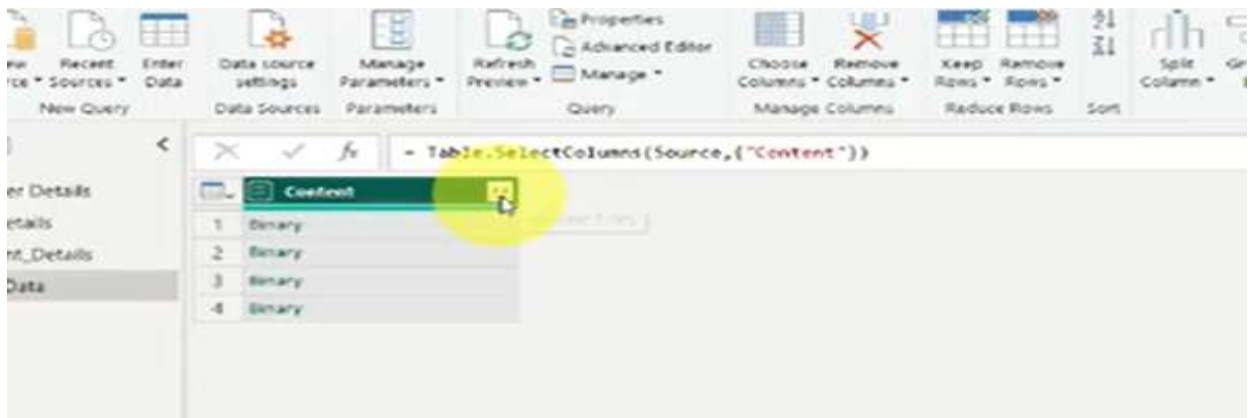


Step 5: now power query window will open which is Etl tool for clean transform and load data. As we don't need other columns than Content column we will remove them.

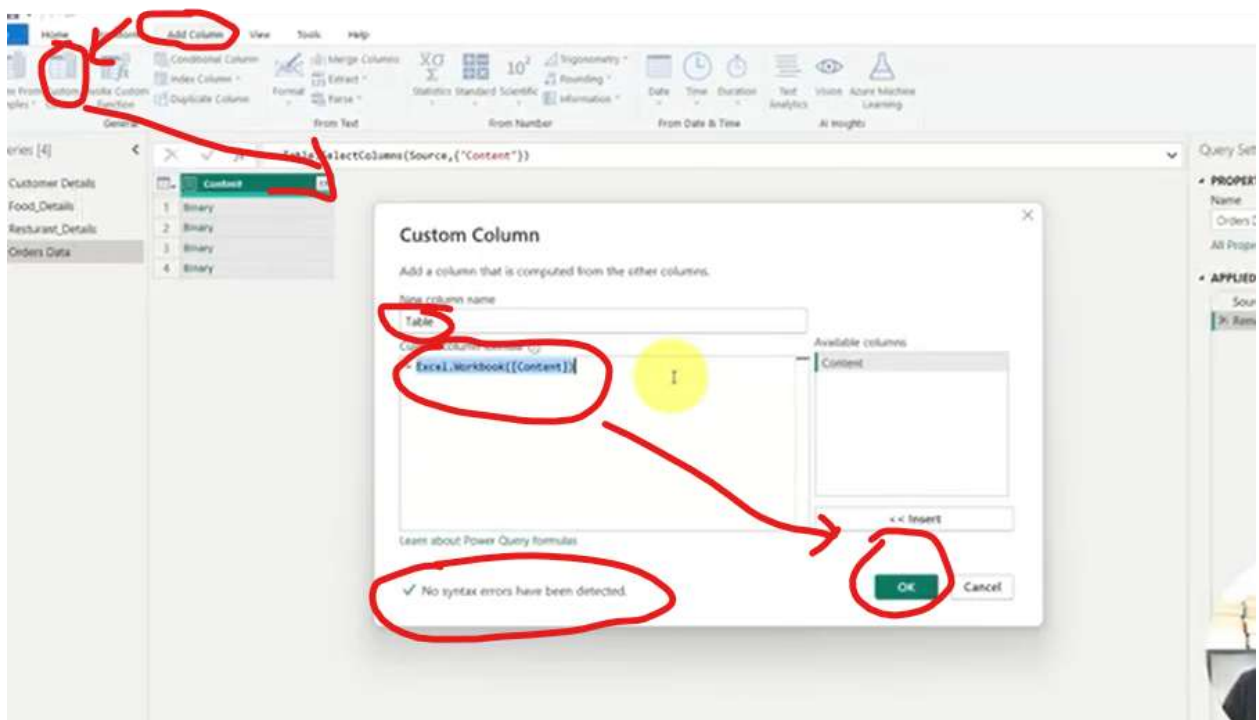


Step 6: select column content and then right click and then click on remove other columns. Hence, in other image we can see columns has been removed.

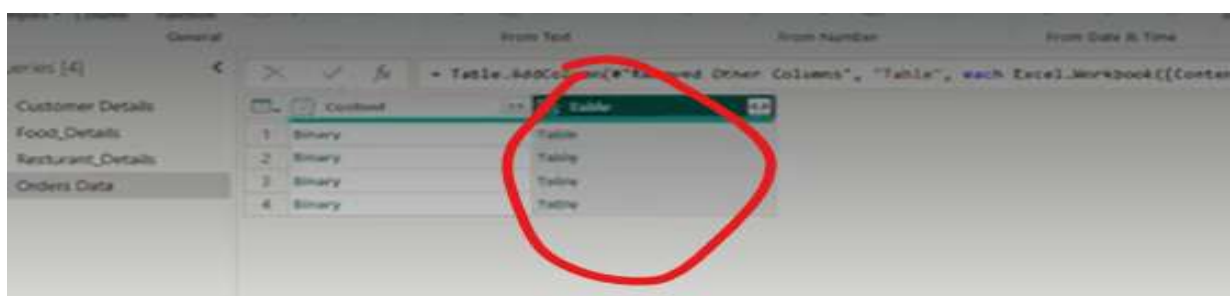




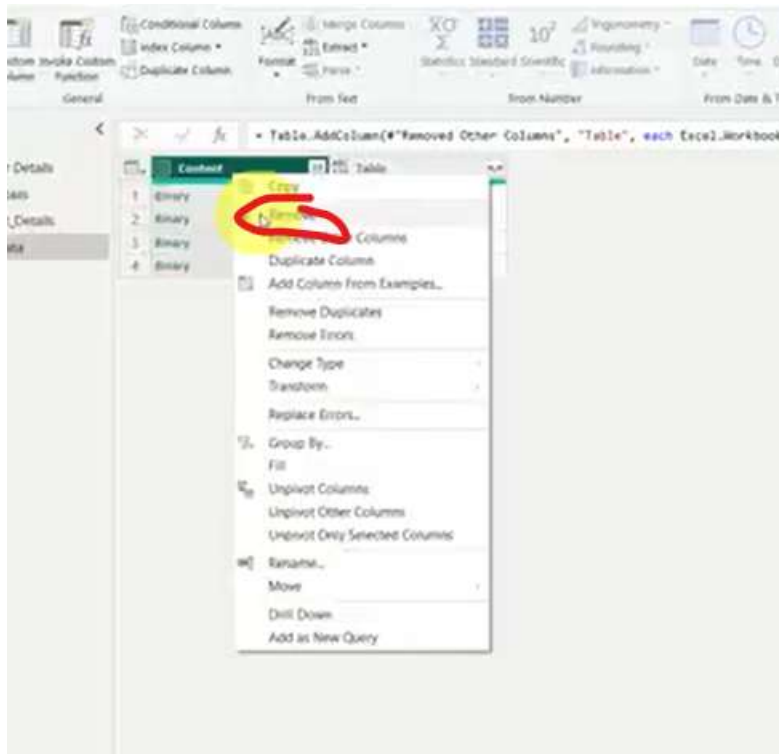
Step 7: now we want extract data from column content so we will go add cloumns → custom data → give column name as table → give formula =Excel.workbook([Content]) . Down there it will show that our formula syntax is correct or not . if correct than click on Ok.



Step 8: now we can see new column Table .



Step 9: we need don't now content column so we will remove right click → remove .



Step 10: now only table column is there now we will expand table column by clicking below icon.

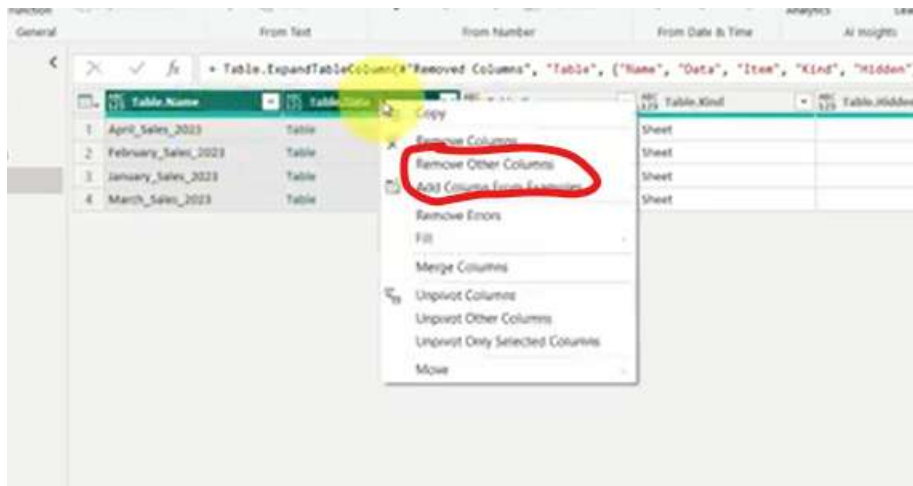


Step 11: now we can see the table column has been expanded but these are not still our data these columns are the properties of sheet

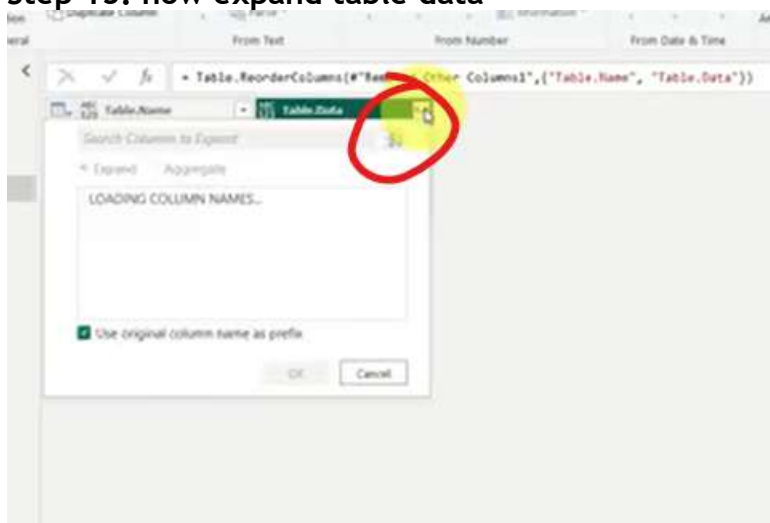


These the sequence workbook will have sheets sheets will have table.
workbook → sheets → table

Step 12: now we only need column table name and table data so will remove other column for that select column table name → ctrl (whole on) than select → remove other columns



Step 13: now expand table data



step 14 : now here's we have table data but the prblm is in 1 row we have header like order_id, etc.

The screenshot shows the Power BI Desktop interface with the expanded table data. The first row contains headers: `order_id`, `orderDate`, `customer_id`, `Restaurant_ID`, and `quantity`. These headers are circled in red. The table contains 17 rows of data, all with the same date (02-04-2023).

Table.Name	order_id	orderDate	customer_id	Restaurant_ID	quantity
April_Sales_2023	256184	02-04-2023	23	211	2142
April_Sales_2023	256185	02-04-2023	24	211	2152
April_Sales_2023	256186	02-04-2023	28	212	2145
April_Sales_2023	256187	02-04-2023	4	211	2149
April_Sales_2023	256188	02-04-2023	30	211	2141
April_Sales_2023	256189	02-04-2023	17	212	2147
April_Sales_2023	256190	02-04-2023	17	211	2142
April_Sales_2023	256191	02-04-2023	15	218	2161
April_Sales_2023	256192	02-04-2023	20	211	2141
April_Sales_2023	256193	02-04-2023	29	212	2147
April_Sales_2023	256194	02-04-2023	25	211	2148
April_Sales_2023	256195	02-04-2023	8	212	2145
April_Sales_2023	256196	02-04-2023	4	211	2150
April_Sales_2023	256197	02-04-2023	13	211	2144
April_Sales_2023	256198	02-04-2023	4	212	2147
April_Sales_2023	256199	02-04-2023	25	217	2165

Step 15: so to solve it we will go to home → select use header as first row

order_id	orderDate	customer_id	Restaurant_ID	FoodItem	quantity
2142	01-04-2023	29	212		2142
2152	01-04-2023	24	212		2152
2143	01-04-2023	28	212		2143
2149	01-04-2023	4	212		2149
2141	01-04-2023	30	211		2141
2147	01-04-2023	17	212		2147
2142	01-04-2023	17	211		2142
2153	01-04-2023	18	218		2153
2141	01-04-2023	39	212		2141
2147	01-04-2023	29	212		2147
2148	01-04-2023	21	212		2148
2149	01-04-2023	5	212		2149
2150	01-04-2023	4	212		2150
2144	01-04-2023	13	211		2144
2147	01-04-2023	4	212		2147
2155	01-04-2023	28	212		2155
2141	01-04-2023	4	212		2141
2142	01-04-2023	28	211		2142
2144	01-04-2023	8	211		2144
2149	01-04-2023	8	212		2149
2141	01-04-2023	29	212		2141
2142	01-04-2023	10	211		2142
2144	01-04-2023	5	211		2144
2149	01-04-2023	8	212		2149
2141	01-04-2023	29	212		2141
2142	01-04-2023	12	211		2142
2147	01-04-2023	18	211		2147

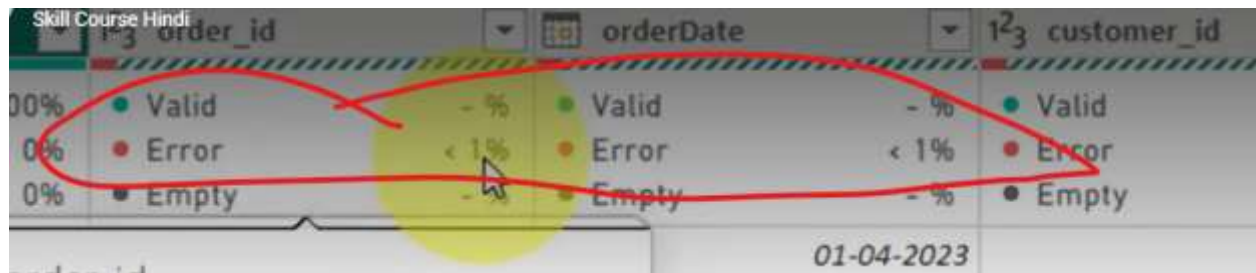
Step 16 : but doing these its showing red lines means error

order_id	orderDate	customer_id	Restaurant_ID	FoodItem	quantity
2142	01-04-2023	29	212		2142
2152	01-04-2023	24	212		2152
2143	01-04-2023	28	212		2143
2149	01-04-2023	4	212		2149
2141	01-04-2023	30	211		2141
2147	01-04-2023	17	212		2147
2142	01-04-2023	17	211		2142
2153	01-04-2023	18	218		2153
2141	01-04-2023	39	212		2141
2147	01-04-2023	29	212		2147
2148	01-04-2023	21	212		2148
2149	01-04-2023	5	212		2149
2150	01-04-2023	4	212		2150
2144	01-04-2023	13	211		2144
2147	01-04-2023	4	212		2147
2155	01-04-2023	28	212		2155
2141	01-04-2023	4	212		2141
2142	01-04-2023	28	211		2142
2144	01-04-2023	8	211		2144
2149	01-04-2023	8	212		2149
2141	01-04-2023	29	212		2141
2142	01-04-2023	10	211		2142
2144	01-04-2023	5	211		2144
2149	01-04-2023	8	212		2149
2141	01-04-2023	29	212		2141
2142	01-04-2023	12	211		2142
2147	01-04-2023	18	211		2147

Step 17 : we can see same from view →column quality

order_id	orderDate	customer_id	Restaurant_ID	FoodItem	quantity	Column Quality
2142	01-04-2023	29	212		2142	Valid
2152	01-04-2023	24	212		2152	Valid
2143	01-04-2023	28	212		2143	Error
2149	01-04-2023	4	212		2149	Valid
2141	01-04-2023	30	211		2141	Valid
2147	01-04-2023	17	212		2147	Valid
2142	01-04-2023	17	211		2142	Valid
2153	01-04-2023	18	218		2153	Valid
2141	01-04-2023	39	212		2141	Valid
2147	01-04-2023	29	212		2147	Valid
2148	01-04-2023	21	212		2148	Valid
2149	01-04-2023	5	212		2149	Valid
2150	01-04-2023	4	212		2150	Valid
2144	01-04-2023	13	211		2144	Valid
2147	01-04-2023	4	212		2147	Valid
2155	01-04-2023	28	212		2155	Valid
2141	01-04-2023	4	212		2141	Valid
2142	01-04-2023	28	211		2142	Valid
2144	01-04-2023	8	211		2144	Valid
2149	01-04-2023	8	212		2149	Valid
2141	01-04-2023	29	212		2141	Valid
2142	01-04-2023	10	211		2142	Valid
2144	01-04-2023	5	211		2144	Valid
2149	01-04-2023	8	212		2149	Valid
2141	01-04-2023	29	212		2141	Valid
2142	01-04-2023	12	211		2142	Valid
2147	01-04-2023	18	211		2147	Valid

Step 18: we can see 1% of error because on 563 row we can see again the header order id, etc the error comes because column type is no. but error is string or text which is invalid .



500	April_Sales_2023	256759	30-04-2023	25	213	2149
501	April_Sales_2023	256760	30-04-2023	5	213	2148
502	April_Sales_2023	256761	30-04-2023	4	213	2155
503	April_Sales_2023	256762	30-04-2023	25	217	2104
504	April_Sales_2023	256763	30-04-2023	20	218	2160
505	February_Sales_2023	256764	01-02-2023	5	212	2148
506	February_Sales_2023	256765	01-02-2023	4	212	2147
507	February_Sales_2023	256766	01-02-2023	25	212	2145
508	February_Sales_2023	256767	01-02-2023	5	212	2147
509	February_Sales_2023	256768	01-02-2023	4	212	2145
510	February_Sales_2023	256769	01-02-2023	29	216	2163
511	February_Sales_2023	256770	01-02-2023	5	212	2147
512	February_Sales_2023	256771	01-02-2023	7	216	2160
513	February_Sales_2023	256772	01-02-2023	20	213	2149

100%	Valid	100%	Valid	100%	Valid
0%	Error	0%	Error	0%	Error
0%	Empty	0%	Empty	0%	Empty
Table.Data.Column1					
1000 (100%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)
Valid	Error	Empty	Valid	Error	Empty
30-04-2023					
30-04-2023					
30-04-2023					
30-04-2023					
30-04-2023					

500	April_Sales_2023	256759	30-04-2023	25	213	2149
501	April_Sales_2023	256760	30-04-2023	5	213	2148
502	April_Sales_2023	256761	30-04-2023	4	213	2155
503	April_Sales_2023	256762	30-04-2023	25	217	2104
504	April_Sales_2023	256763	30-04-2023	20	218	2160
505	February_Sales_2023	256764	01-02-2023	5	212	2148
506	February_Sales_2023	256765	01-02-2023	4	212	2147
507	February_Sales_2023	256766	01-02-2023	25	212	2145
508	February_Sales_2023	256767	01-02-2023	5	212	2147
509	February_Sales_2023	256768	01-02-2023	4	212	2145
510	February_Sales_2023	256769	01-02-2023	29	216	2163
511	February_Sales_2023	256770	01-02-2023	5	212	2147
512	February_Sales_2023	256771	01-02-2023	7	216	2160
513	February_Sales_2023	256772	01-02-2023	20	213	2149

Step 19: so now we will remove error select columns → right click → remove errors other way is that click on error and click on remove errors.

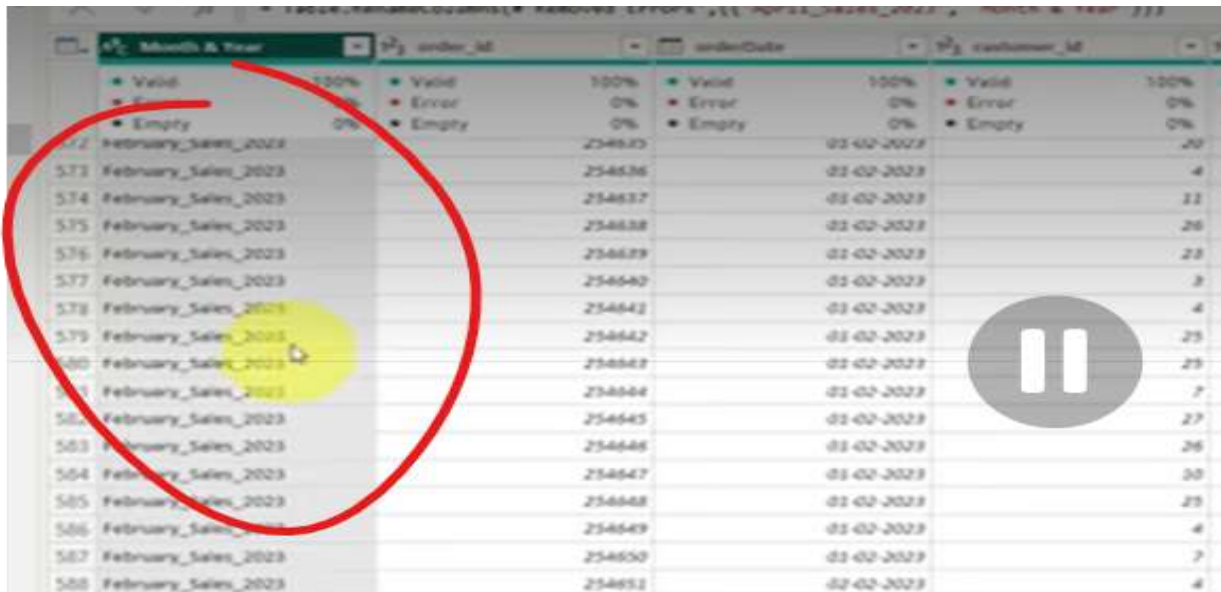
Table.RenameColumns(# ...

ABC Month & Year	123 order_id
Valid 100%	Valid
Error 0%	Error
Empty 0%	Empty
1 April_Sales_2023	

Step 22: now we see that our table is append first its table of April than of februray and than continue .

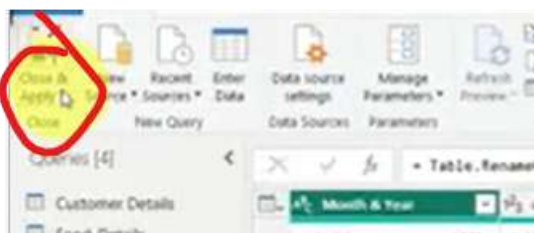
Table.RenameColumns(#"Removed Errors",{{"April_Sales_2023", "Month & Year"}})

Month & Year	order_id	orderDate	customer_id	Restaurant_ID
Valid 100%	Valid 100%	Valid 100%	Valid 100%	Valid 100%
Error 0%	Error 0%	Error 0%	Error 0%	Error 0%
Empty 0%	Empty 0%	Empty 0%	Empty 0%	Empty 0%
174 April_Sales_2023	256337	23-04-2023	25	
175 April_Sales_2023	256338	23-04-2023	25	
176 April_Sales_2023	256339	23-04-2023	3	
177 April_Sales_2023	256360	23-04-2023	25	
178 April_Sales_2023	256362	24-04-2023	8	
179 April_Sales_2023	256362	24-04-2023	8	
180 April_Sales_2023	256363	24-04-2023	22	
181 April_Sales_2023	256364	24-04-2023	4	
182 April_Sales_2023	256365	24-04-2023	9	
183 April_Sales_2023	256366	24-04-2023	25	
184 April_Sales_2023	256367	24-04-2023	1	
185 April_Sales_2023	256368	24-04-2023	7	
186 April_Sales_2023	256369	24-04-2023	29	
187 April_Sales_2023	256370	24-04-2023	7	
188 April_Sales_2023	256372	24-04-2023	4	
189 April_Sales_2023	256372	24-04-2023	30	
190 April_Sales_2023	256373	24-04-2023	5	
191 April_Sales_2023	256374	24-04-2023	30	
192 April_Sales_2023	256375	24-04-2023	9	
193 April_Sales_2023	256376	24-04-2023	29	
194 April_Sales_2023	256377	24-04-2023	22	
195 April_Sales_2023	256378	24-04-2023	7	
196 April_Sales_2023	256379	24-04-2023	27	



Month & Year	order_id	orderDate	customer_id
February_Sales_2023	254632	01-02-2023	20
February_Sales_2023	254636	01-02-2023	4
February_Sales_2023	254637	01-02-2023	11
February_Sales_2023	254638	01-02-2023	26
February_Sales_2023	254639	01-02-2023	23
February_Sales_2023	254640	01-02-2023	3
February_Sales_2023	254641	01-02-2023	4
February_Sales_2023	254642	01-02-2023	25
February_Sales_2023	254643	01-02-2023	25
February_Sales_2023	254644	01-02-2023	7
February_Sales_2023	254645	01-02-2023	27
February_Sales_2023	254646	01-02-2023	26
February_Sales_2023	254647	01-02-2023	30
February_Sales_2023	254648	01-02-2023	25
February_Sales_2023	254649	01-02-2023	4
February_Sales_2023	254650	01-02-2023	7
February_Sales_2023	254651	01-02-2023	4

Step 23: now will click on close and apply these will load our order data in power BI.



Step 24 : we can that order data has been loaded in the power query .



Q1. What is Append in Power BI?

Answer:

Append is used to combine two or more tables by adding rows from one table below another table.

Q2. What is the difference between Append and Merge?

Append

Merge

Adds Rows

Adds Columns

Append

Merge

Vertical Combination

Horizontal Combination

Increases Records

Increases Fields

Answer:

Append adds rows, whereas Merge adds columns based on a common column.

Q3. What is the condition for Append?

Answer:

The tables should have the same or similar column names and data structure.

Q4. Why do we use Folder as a data source?

Answer:

Folder data source is used when multiple files need to be combined automatically into a single table.

Q5. What does Content column contain?

Answer:

The Content column contains binary data of the files.

Q6. Why do we use Excel.Workbook([Content])?

Answer:

It is used to extract Excel sheet and table data from the binary Content column.

Q7. What is Binary Data?

Answer:

Binary data is the raw file data stored in Power BI before extracting its contents.

Q8. Why do we use Remove Other Columns?

Answer:

To keep only the required columns and remove unnecessary columns.

Q9. What is the purpose of Expand Column?

Answer:

Expand is used to extract data stored inside nested tables, records, or lists.

Q10. Explain Workbook → Sheet → Table.

Answer:

- Workbook = Excel File
 - Sheet = Worksheet inside Excel
 - Table = Actual data inside the sheet
-

Q11. Why do we use Use First Row as Headers?

Answer:

To convert the first row values into column names.

Q12. Why do errors occur after promoting headers?

Answer:

Errors occur when a column's data type does not match the actual data value.

Example:

A Number column contains text like "Order ID".

Q13. How can you identify errors in Power Query?

Answer:

Using:

- Column Quality
- Column Distribution
- Column Profile

under the View tab.

Q14. How do you remove errors in Power Query?

Answer:
Right-click the column and select Remove Errors.

Q15. What is Column Quality?

Answer:
Column Quality shows:

- Valid values
- Error values
- Empty values

in percentage format.

Q16. What happens when you click Close & Apply?

Answer:
The transformed data is loaded from Power Query into Power BI.

Most Important Interview Question

Q17. Why did we use Append in this project?

Answer:
We had multiple monthly order files (April, February, etc.) stored in a folder. Using Append, we combined all monthly data into a single Orders table for reporting and analysis.